

## B1 Fluids — Model Solutions, Trinity Term 2022

### Question 1 — Darcy flow

**Problem.** Stack of  $n$  rigid sheets, gap  $b \ll w, L$ , fluid  $(\rho, \eta)$ , pressure drop  $\Delta p$  across length  $L$ . Derive velocity field and Darcy permeability  $\Pi_0$ ; comment on inertia. Then generalise, prove  $\nabla \cdot (\rho \mathbf{q}) = 0$ , solve a spherical injection, compare two aquifers, and discuss gravity.

(a)

Take  $\hat{\mathbf{x}}$  along the plates,  $z \in [0, b]$  across the gap. Symmetry and no-penetration:

$$\mathbf{u} = u_x(z)\hat{\mathbf{x}}, \quad u_y = u_z = 0.$$

Steady, unidirectional,  $z$ -independent pressure: Navier–Stokes reduces to:

$$0 = -\frac{\partial p}{\partial x} + \eta \frac{\partial^2 u_x}{\partial z^2}, \quad \frac{\partial p}{\partial z} = 0.$$

Constant pressure gradient:

$$\frac{\partial p}{\partial x} = -\frac{\Delta p}{L}.$$

Integrate twice in  $z$ :

$$u_x(z) = -\frac{1}{2\eta} \frac{\Delta p}{L} z(b-z) \cdot (-1) + C_1 z + C_2.$$

Apply no-slip  $u_x(0) = u_x(b) = 0$ :

$$u_x(z) = \frac{\Delta p}{2\eta L} z(b-z).$$

Volume flux per unit cross-sectional area:

$$q = \frac{1}{b} \int_0^b u_x \, dz.$$

Evaluate the integral:

$$q = \frac{\Delta p}{2\eta L b} \cdot \frac{b^3}{6} = \frac{b^2}{12\eta} \frac{\Delta p}{L}.$$

Compare with  $q = (\Pi_0/\eta)(\Delta p/L)$ :

$$\boxed{\Pi_0 = \frac{b^2}{12}}.$$

**Inertia.** For unidirectional  $u_x(z)$ :  $(\mathbf{u} \cdot \nabla)\mathbf{u} = u_x \partial_x u_x \hat{\mathbf{x}} = 0$  identically. Force balance is pressure  $\leftrightarrow$  viscous stress only; inertia plays no role. Equivalently  $\text{Re} = Ub/\nu \ll 1$  in the Darcy regime.

(b)

Mass conservation in fixed volume  $V$ :

$$\frac{d}{dt} \int_V \rho \, dV = - \oint_{\partial V} \rho \mathbf{q} \cdot d\mathbf{S}.$$

Steady, rigid matrix: LHS = 0. Apply divergence theorem:

$$\int_V \nabla \cdot (\rho \mathbf{q}) \, dV = 0.$$

Volume arbitrary, integrand continuous:

$$\boxed{\nabla \cdot (\rho \mathbf{q}) = 0.}$$

(c)

Spherically symmetric,  $\rho$  constant,  $\nabla \cdot \mathbf{q} = 0$  with  $\mathbf{q} = q_r(r) \hat{\mathbf{r}}$ :

$$\frac{1}{r^2} \frac{d}{dr} (r^2 q_r) = 0.$$

Integrate:

$$q_r = \frac{C}{r^2}.$$

Darcy without gravity:  $\mathbf{q} = -(\Pi_1/\eta) \nabla p$ , so  $q_r = -(\Pi_1/\eta) \, dp/dr$ :

$$\frac{dp}{dr} = -\frac{\eta C}{\Pi_1 r^2}.$$

Integrate from  $r$  to  $\infty$  with  $p(\infty) = p_0$ :

$$p(r) - p_0 = \frac{\eta C}{\Pi_1 r}.$$

Apply  $p(a) = p_0 + \Delta p$ :

$$C = \frac{\Pi_1 \Delta p a}{\eta}.$$

Hence:

$$\boxed{q_r(r) = \frac{\Pi_1 \Delta p a}{\eta r^2}.$$

**Aquifer choice.** Pump power at fixed flow rate  $Q = 4\pi a^2 q_r(a)$ :

$$Q = \frac{4\pi a \Pi_1 \Delta p}{\eta}.$$

Invert for  $\Delta p$ :

$$\Delta p = \frac{Q \eta}{4\pi a \Pi_1}.$$

Power dissipated:

$$\mathcal{P} = Q \Delta p = \frac{Q^2 \eta}{4\pi a \Pi_1}.$$

Carman–Kozeny scaling (cf. part (a) where  $\Pi_0 = b^2/12$ ):

$$\Pi_1 \propto d^2.$$

Use  $d_A = 2d_B$ :

$$\Pi_{1,A} = 4\Pi_{1,B}.$$

Use  $\eta_A = 2\eta_B$ :

$$\frac{\mathcal{P}_A}{\mathcal{P}_B} = \frac{\eta_A/\Pi_{1,A}}{\eta_B/\Pi_{1,B}} = \frac{2}{4} = \frac{1}{2}.$$

Aquifer A: half the pumping power at fixed  $Q$ .

Assumptions: same source radius  $a$ , Darcy regime ( $\text{Re} \ll 1$ ),  $\Pi \propto d^2$ , equal densities (gravity drops out).

(d)

Reinstate gravity, define piezometric pressure:

$$P \equiv p + \rho g z.$$

Darcy with gravity:

$$\frac{\eta}{\Pi_1} \mathbf{q} = -\nabla p - \rho g \hat{\mathbf{z}} = -\nabla P.$$

Far-field hydrostatic:  $P_\infty = p_0(0)$  constant. Source:  $P(a) \approx p_0(0) + \Delta p$  (for  $a$  small). The problem in  $P$  is identical to part (c):

$$q_r(r) = \frac{\Pi_1 \Delta p a}{\eta r^2} \text{ unchanged.}$$

For matched densities gravity only relabels the absolute pressure. Density mismatch  $\Delta\rho$  would break spherical symmetry (buoyant plume).

## Question 2 — Stagnation-point boundary layer

**Problem.** Inviscid stagnation flow  $\psi = -Axy$  above wall  $y = 0$ ,  $A > 0$ , gravity  $-g\hat{\mathbf{y}}$ . Distinguish Eulerian/Lagrangian acceleration; verify wall BC; find  $p_\infty$ ; derive boundary-layer (BL) equations; reduce via  $\psi = -Ax\delta f(\eta)$ ; estimate  $\delta$  and turbulence onset.

(a)

Eulerian acceleration:  $\partial \mathbf{u} / \partial t$  at fixed point (river-bank probe). Lagrangian acceleration:  $D\mathbf{u} / Dt = \partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla) \mathbf{u}$  following a parcel (drifting leaf). Steady river:  $\partial_t \mathbf{u} = 0$  but parcel still accelerates through rapids via  $(\mathbf{u} \cdot \nabla) \mathbf{u}$ . Newton's law governs the Lagrangian acceleration.

(b)

Velocity from stream function:

$$v_x = \frac{\partial \psi}{\partial y} = -Ax, \quad v_y = -\frac{\partial \psi}{\partial x} = Ay.$$

Inviscid wall BC is no-penetration only:

$$v_y(x, 0) = 0 \quad \checkmark.$$

Steady irrotational flow: Bernoulli  $p + \frac{1}{2}\rho|\mathbf{v}|^2 + \rho gy = \text{const.}$  Use  $|\mathbf{v}|^2 = A^2(x^2 + y^2)$ :

$$p_\infty(x, y) = p_0 - \frac{1}{2}\rho A^2(x^2 + y^2) - \rho gy.$$

Streamwise gradient:

$$\frac{\partial p_\infty}{\partial x} = -\rho A^2 x.$$

(c)

At finite Re the no-slip condition  $v_x(x, 0) = 0$  replaces inviscid free-slip. Outer flow gives  $v_x(x, 0) = -Ax \neq 0$ : incompatible. Viscosity reasserts itself in a thin layer of thickness  $\delta(x) \ll L$ .

Scales:  $L \sim x$ ,  $\delta \ll L$ . Incompressibility:

$$\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0.$$

Order-of-magnitude:

$$\frac{V_x}{L} \sim \frac{V_y}{\delta}.$$

Solve for  $V_y$ :

$$V_y \sim \frac{\delta}{L} V_x \ll V_x.$$

$x$ -momentum (steady, incompressible):

$$\mathbf{v} \cdot \nabla v_x = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left( \frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} \right).$$

Compare viscous terms:  $\nu V_x/L^2$  vs.  $\nu V_x/\delta^2$ :

$$\frac{\partial^2 v_x}{\partial x^2} \ll \frac{\partial^2 v_x}{\partial y^2}.$$

Convective  $\sim$  dominant viscous balance:

$$\frac{V_x^2}{L} \sim \frac{\nu V_x}{\delta^2}.$$

Solve for  $\delta$ :

$$\delta \sim \sqrt{\frac{\nu L}{V_x}} = L \text{Re}^{-1/2} \ll L.$$

$y$ -momentum: each term  $\lesssim (\delta/L)V_x^2/L$ ; at leading order:

$$\frac{\partial p}{\partial y} \approx 0.$$

Pressure across layer matches outer:

$$\frac{\partial p}{\partial x} = \frac{\partial p_\infty}{\partial x}, \quad \frac{\partial p}{\partial y} = \frac{\partial p_\infty}{\partial y}.$$

$$\mathbf{v} \cdot \nabla v_x = \nu \frac{\partial^2 v_x}{\partial y^2} - \frac{1}{\rho} \frac{\partial p_\infty}{\partial x}, \quad \frac{\partial p}{\partial y} = \frac{\partial p_\infty}{\partial y}, \quad \nabla \cdot \mathbf{v} = 0.$$

(d)

Try  $\psi = -Ax\delta f(\eta)$ ,  $\eta = y/\delta$ ,  $\delta = \text{const}$ :

$$v_x = \frac{\partial \psi}{\partial y} = -Ax f'(\eta), \quad v_y = -\frac{\partial \psi}{\partial x} = A\delta f(\eta).$$

Convective term:

$$\mathbf{v} \cdot \nabla v_x = v_x \frac{\partial v_x}{\partial x} + v_y \frac{\partial v_x}{\partial y}.$$

Substitute:

$$\mathbf{v} \cdot \nabla v_x = (-Ax f')(-A f') + (A\delta f) \left( -\frac{Ax}{\delta} f'' \right).$$

Simplify:

$$\mathbf{v} \cdot \nabla v_x = A^2 x [(f')^2 - f f''].$$

Viscous term:

$$\nu \frac{\partial^2 v_x}{\partial y^2} = -\frac{\nu Ax}{\delta^2} f'''.$$

Pressure gradient:  $-\rho^{-1} \partial_x p_\infty = A^2 x$ . Substitute into BL equation:

$$A^2 x [(f')^2 - f f''] = -\frac{\nu Ax}{\delta^2} f''' + A^2 x.$$

Divide through by  $A^2 x$ :

$$(f')^2 - f f'' = -\frac{\nu}{A\delta^2} f''' + 1.$$

Choose  $\delta$  to set viscous coefficient to unity:

$$\delta = \sqrt{\frac{\nu}{A}}.$$

Rearrange:

$$\boxed{f''' + f f'' - (f')^2 = -1.}$$

Boundary conditions (no-penetration, no-slip, far-field match):

$$\boxed{f(0) = 0, \quad f'(0) = 0, \quad f'(\eta) \rightarrow 1 \text{ as } \eta \rightarrow \infty.}$$

(e)

From part (d):

$$\boxed{\delta = \sqrt{\nu/A} \text{ (independent of } x, y, \rho, p_0\text{).}}$$

Controlled by viscous diffusion rate  $\nu$  vs. strain  $A$ ; the linear acceleration  $-Ax$  exactly compensates the longer transit time at large  $x$ , giving a uniform-thickness layer.

**Turbulence onset.** Local Reynolds number on length  $x$ :

$$\text{Re}_x = \frac{|v_x| \delta}{\nu}.$$

Substitute  $|v_x| = Ax$ ,  $\delta = \sqrt{\nu/A}$ :

$$\text{Re}_x = \frac{Ax \sqrt{\nu/A}}{\nu} = x \sqrt{\frac{A}{\nu}}.$$

Set transition  $\text{Re}_x \sim 10^3$ :

$$\boxed{l \sim 10^3 \sqrt{\nu/A}.$$

### Question 3 — Rayleigh–Taylor in a cylindrical tank

**Problem.** Closed cylinder  $0 < r < a$ ,  $-h < z < h$ . Freshwater ( $\rho_1$ ) below, saltwater ( $\rho_2 > \rho_1$ ) above, interface  $z = \eta(r, t)$ , no surface tension,  $\mathbf{g} = -g\hat{\mathbf{z}}$ , no diffusion. Membrane bursts at  $t = 0$ . (a) Derive Laplace + linearised BCs; (b) normal modes; (c) baroclinic vorticity equation and physical mechanism.

(a)

Initial vorticity zero in each layer; uniform  $\rho_i$ , no viscosity, no baroclinic source. Kelvin's theorem: vorticity stays zero, flow irrotational:

$$\mathbf{v}_i = \nabla \phi_i.$$

Incompressibility:

$$\nabla^2 \phi_i = 0 \quad (i = 1, 2).$$

**Boundary conditions.** Impermeable walls:

$$\left. \frac{\partial \phi_1}{\partial z} \right|_{z=-h} = 0, \quad \left. \frac{\partial \phi_2}{\partial z} \right|_{z=h} = 0, \quad \left. \frac{\partial \phi_i}{\partial r} \right|_{r=a} = 0.$$

Kinematic at  $z = \eta$ :  $D(z - \eta)/Dt = 0$ :

$$\frac{\partial \eta}{\partial t} + v_r \frac{\partial \eta}{\partial r} = v_z.$$

Linearise (drop quadratic, evaluate at  $z = 0$ ):

$$\left. \frac{\partial \eta}{\partial t} = \frac{\partial \phi_i}{\partial z} \right|_{z=0} \quad (i = 1, 2).$$

Linearised Bernoulli in each layer:

$$p_i = -\rho_i \frac{\partial \phi_i}{\partial t} - \rho_i g z.$$

Continuity of pressure  $p_1 = p_2$  at  $z = \eta$ , evaluate at  $z = 0$  to first order:

$$\rho_2 \frac{\partial \phi_2}{\partial t} - \rho_1 \frac{\partial \phi_1}{\partial t} = (\rho_1 - \rho_2) g \eta \quad \text{at } z = 0.$$

(b)

Modal ansatz consistent with bounded axisymmetric Laplacian solution:

$$\phi_i = J_0(\alpha r) [A_i e^{\alpha z} + B_i e^{-\alpha z}] e^{\sigma t}, \quad \eta = \eta_0 J_0(\alpha r) e^{\sigma t}.$$

Apply radial wall BC, using  $J'_0 = -J_1$ :

$$J_1(\alpha a) = 0.$$

Hence:

$$\alpha_n a = j_{1,n}, \quad n = 1, 2, 3, \dots$$

Apply bottom wall  $\partial_z \phi_1|_{-h} = 0$ :

$$\phi_1 = C_1 \cosh[\alpha_n(z + h)] J_0(\alpha_n r) e^{\sigma t}.$$

Apply top wall  $\partial_z \phi_2|_h = 0$ :

$$\phi_2 = C_2 \cosh[\alpha_n(z - h)] J_0(\alpha_n r) e^{\sigma t}.$$

Kinematic at  $z = 0$  for layer 1:

$$\sigma \eta_0 = \alpha_n C_1 \sinh(\alpha_n h).$$

Kinematic at  $z = 0$  for layer 2:

$$\sigma \eta_0 = -\alpha_n C_2 \sinh(\alpha_n h).$$

Solve for amplitudes:

$$C_1 = \frac{\sigma \eta_0}{\alpha_n \sinh(\alpha_n h)}, \quad C_2 = -\frac{\sigma \eta_0}{\alpha_n \sinh(\alpha_n h)}.$$

Insert into dynamic condition at  $z = 0$  (using  $\cosh(-x) = \cosh x$ ):

$$\rho_2 \sigma C_2 \cosh(\alpha_n h) - \rho_1 \sigma C_1 \cosh(\alpha_n h) = (\rho_1 - \rho_2) g \eta_0.$$

Substitute  $C_1, C_2$ :

$$-\frac{\sigma^2 \eta_0 (\rho_1 + \rho_2) \cosh(\alpha_n h)}{\alpha_n \sinh(\alpha_n h)} = (\rho_1 - \rho_2) g \eta_0.$$

Divide by  $\eta_0$  and rearrange:

$$\sigma_n^2 = \left( \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} \right) g \alpha_n \tanh(\alpha_n h).$$

**Stability.**  $\rho_2 > \rho_1 \Rightarrow \sigma_n^2 > 0 \Rightarrow \sigma_n$  real, one root positive: *unconditionally unstable*. Limits:

$$\alpha_n h \ll 1 : \quad \sigma_n^2 \approx \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} g \alpha_n^2 h,$$

$$\alpha_n h \gg 1 : \quad \sigma_n^2 \approx \frac{\rho_2 - \rho_1}{\rho_2 + \rho_1} g \alpha_n.$$

Growth rate increases monotonically with  $\alpha_n$ ; without surface tension or viscosity the highest geometrically allowed mode dominates.

(c)

Variable-density momentum equation:

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = \frac{\mathbf{F}}{\rho}.$$

Take curl, use 2D identity  $\nabla \times [(\mathbf{v} \cdot \nabla) \mathbf{v}] = (\mathbf{v} \cdot \nabla) \omega$  for  $\nabla \cdot \mathbf{v} = 0$ :

$$\frac{\partial \omega}{\partial t} + \mathbf{v} \cdot \nabla \omega = \nabla \times \left( \frac{\mathbf{F}}{\rho} \right).$$

Inviscid:  $\mathbf{F} = -\nabla p + \rho \mathbf{g}$ . Body-force curl:

$$\nabla \times \mathbf{g} = 0.$$

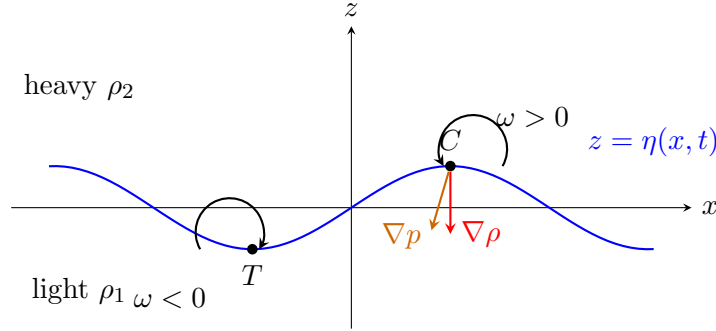
Pressure-gradient curl:

$$\nabla \times \left( -\frac{\nabla p}{\rho} \right) = -\nabla \left( \frac{1}{\rho} \right) \times \nabla p.$$

Use  $\nabla(1/\rho) = -\nabla \rho / \rho^2$ :

$$\frac{\partial \omega}{\partial t} + \mathbf{v} \cdot \nabla \omega = \frac{\nabla \rho \times \nabla p}{\rho^2}.$$

**Mechanism (sketch below).** Within the diffuse interface of width  $\delta$ :  $\nabla\rho$  is large and predominantly vertical (downward into the light fluid);  $\nabla p \approx -\rho g \hat{\mathbf{z}}$  but tilts horizontally where the interface is perturbed. The cross product  $\nabla\rho \times \nabla p$  alternates sign along the wave: positive vorticity on one flank of each crest, negative on the other. The induced circulation lifts light fluid into crests and pushes heavy fluid into troughs, *reinforcing* the perturbation: exponential growth.



**Variation of  $\sigma_n$  with  $n$ .** Energy argument: KE  $\sim (\rho_1 + \rho_2)v^2/(2k_n)$ ; PE released  $\sim (\rho_2 - \rho_1)g\eta^2/2$ . Equate, with  $v \sim \sigma\eta$ :

$$\sigma_n^2 \sim \frac{\rho_2 - \rho_1}{\rho_1 + \rho_2} g k_n.$$

Short waves (large  $n$ ) localise motion at the interface where the baroclinic torque acts: faster growth. Finite depth  $h$  caps the inertia at  $\sim h$  once  $k_n h \lesssim 1$ , suppressing  $\sigma_n$  by  $\tanh(k_n h)$ .

## Question 4 — Stokes flow past a clean bubble

**Problem.** (a) Derive  $\nabla^2 p, \nabla^2 \boldsymbol{\omega} \approx 0$  in Stokes flow; (b) verify incompressibility of axisymmetric streamfunction and write BCs at the bubble; (c) verify  $\Psi = Cr \sin^2 \theta$  and obtain drag  $4\pi\eta a U$ ; (d) shrinking bubble at terminal velocity, find  $a(z)$  and critical depth  $d$  for  $a_0 = 10^{-4}$  m.

(a)

Low-Re conditions:  $\text{Re} = UL/\nu \ll 1$ ; small  $U$ , small  $L$ , or large  $\nu$ . Examples: bacterial swimming, microfluidics, lava/honey flow.

Drop inertia in Navier–Stokes:

$$0 \approx -\nabla p + \eta \nabla^2 \mathbf{u} + \rho \mathbf{g}.$$

Absorb gravity into modified pressure  $p \rightarrow p + \rho g z$ . Take divergence, use  $\nabla \cdot \mathbf{u} = 0$ :

$$\boxed{\nabla^2 p \approx 0.}$$

Take curl, use  $\nabla \times \nabla p = 0$  and  $\nabla \times \nabla^2 \mathbf{u} = \nabla^2 \boldsymbol{\omega}$ :

$$\boxed{\nabla^2 \boldsymbol{\omega} \approx 0.}$$

(b)

Axisymmetric divergence:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2} \frac{\partial(r^2 v_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta v_\theta)}{\partial \theta}.$$

Substitute streamfunction definitions:

$$\nabla \cdot \mathbf{v} = \frac{1}{r^2 \sin \theta} \frac{\partial^2 \Psi}{\partial r \partial \theta} - \frac{1}{r^2 \sin \theta} \frac{\partial^2 \Psi}{\partial \theta \partial r}.$$

Schwarz cancellation:

$$\nabla \cdot \mathbf{v} = 0 \checkmark.$$

**BCs at the bubble.** No-penetration at  $r = a$  (bubble-rest frame, far field  $-U\hat{\mathbf{z}}$ ):

$$v_r(a, \theta) = 0, \quad v_r \rightarrow -U \cos \theta, \quad v_\theta \rightarrow U \sin \theta \text{ as } r \rightarrow \infty.$$

Inviscid interior  $\Rightarrow$  tangential viscous stress vanishes:

$$\sigma_{r\theta}|_{r=a} = 2\eta e_{r\theta}|_{r=a} = 0.$$

Use  $v_r(a, \theta) = 0 \Rightarrow \partial v_r / \partial \theta|_a = 0$ :

$$\boxed{\left. \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) \right|_{r=a} = 0.}$$

(c)

Full streamfunction (uniform far-field plus given correction):

$$\Psi = -\frac{1}{2} U r^2 \sin^2 \theta + C r \sin^2 \theta.$$

Compute  $v_r$ :

$$v_r = \frac{1}{r^2 \sin \theta} \frac{\partial \Psi}{\partial \theta} = -U \cos \theta + \frac{2C \cos \theta}{r}.$$

Compute  $v_\theta$ :

$$v_\theta = -\frac{1}{r \sin \theta} \frac{\partial \Psi}{\partial r} = U \sin \theta - \frac{C \sin \theta}{r}.$$

Apply  $v_r(a, \theta) = 0$ :

$$C = \frac{Ua}{2}.$$

Form  $v_\theta/r$  and differentiate:

$$\frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) = -\frac{U \sin \theta}{r^2} + \frac{2C \sin \theta}{r^3}.$$

Evaluate at  $r = a$  with  $C = Ua/2$ :

$$\left. \frac{\partial}{\partial r} \left( \frac{v_\theta}{r} \right) \right|_{r=a} = -\frac{U \sin \theta}{a^2} + \frac{U \sin \theta}{a^2} = 0 \checkmark.$$

Far-field matching at  $r \rightarrow \infty$ :

$$v_r \rightarrow -U \cos \theta \checkmark.$$

**Drag.**  $z$ -component of surface stress:

$$T_z = (-p + 2\eta e_{rr}) \cos \theta - 2\eta e_{r\theta} \sin \theta.$$

Tangential stress vanishes at  $r = a$ :

$$2\eta e_{r\theta}|_{r=a} = 0.$$

Compute  $\partial v_r / \partial r$  at  $r = a$  using  $C = Ua/2$ :

$$\left. \frac{\partial v_r}{\partial r} \right|_{r=a} = -\frac{2C \cos \theta}{a^2} = -\frac{U \cos \theta}{a}.$$

Hence radial viscous stress at the surface:

$$2\eta e_{rr}|_{r=a} = -\frac{2\eta U \cos \theta}{a}.$$

Pressure on the surface (given form,  $r = a$ ):

$$p|_{r=a} = p_0 + \frac{\eta U \cos \theta}{a}.$$

Combine:

$$T_z|_{r=a} = -p_0 \cos \theta - \frac{3\eta U \cos^2 \theta}{a}.$$

Integrate over the sphere:

$$F = 2\pi a^2 \int_0^\pi \left[ -p_0 \cos \theta - \frac{3\eta U \cos^2 \theta}{a} \right] \sin \theta \, d\theta.$$

$$\int_0^\pi \cos \theta \sin \theta \, d\theta = 0:$$

$$F = -2\pi a^2 \cdot \frac{3\eta U}{a} \cdot \int_0^\pi \cos^2 \theta \sin \theta \, d\theta.$$

Evaluate  $\int_0^\pi \cos^2 \theta \sin \theta \, d\theta = 2/3$ :

$$F = -4\pi \eta a U.$$

Magnitude:

$$\boxed{|F| = 4\pi \eta a U.}$$

(2/3 of Stokes drag on rigid sphere; tangential stress missing at the free surface.)

**(d)**

Terminal velocity: buoyancy = drag (with  $\rho_a \ll \rho_w$ ):

$$\frac{4}{3}\pi a^3 \rho_w g = 4\pi \eta_w a U.$$

Solve:

$$U(a) = \frac{\rho_w g a^2}{3\eta_w}.$$

Mass:  $M = \frac{4}{3}\pi a^3 \rho_a$ , so:

$$\frac{dM}{dt} = 4\pi a^2 \rho_a \frac{da}{dt}.$$

Equate to dissolution rate  $-4\pi a D_a \rho_a$ :

$$a \frac{da}{dt} = -D_a.$$

Convert to  $z$  via  $dz/dt = U$ :

$$a U(a) \frac{da}{dz} = -D_a.$$

Substitute  $U(a)$ :

$$\frac{\rho_w g}{3\eta_w} a^3 \frac{da}{dz} = -D_a.$$

Separate and integrate from  $a_0$  at  $z = 0$ :

$$\frac{\rho_w g}{3\eta_w} \int_{a_0}^a a'^3 da' = -D_a z.$$

Evaluate:

$$\frac{\rho_w g}{12\eta_w} (a^4 - a_0^4) = -D_a z.$$

Rearrange:

$$a^4(z) = a_0^4 - \frac{12\eta_w D_a}{\rho_w g} z.$$

**Critical depth.** Set  $a(d) = 0$ :

$$d = \frac{\rho_w g a_0^4}{12 \eta_w D_a}.$$

Numerical evaluation:  $a_0 = 10^{-4} \text{ m}$ ,  $\rho_w = 10^3 \text{ kg m}^{-3}$ ,  $g = 9.81 \text{ m s}^{-2}$ ,  $\eta_w = 10^{-3} \text{ Pa s}$ ,  $D_a = 10^{-10} \text{ m}^2 \text{ s}^{-1}$ .

Numerator:

$$\rho_w g a_0^4 = (10^3)(9.81)(10^{-4})^4 = 9.81 \times 10^{-13} \text{ kg m}^{-1} \text{ s}^{-2} \cdot \text{m}^4.$$

Denominator:

$$12 \eta_w D_a = 12 (10^{-3})(10^{-10}) = 1.2 \times 10^{-12}.$$

Divide:

$$d = \frac{9.81 \times 10^{-13}}{1.2 \times 10^{-12}} \text{ m}.$$

$$d \approx 0.82 \text{ m}.$$